

Supplementary Material:

Hamiltonian-Inspired Energy Dissipation and Chunk-Wise Variational State Coupling for Efficient Multimodal State-Space Models

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I. OVERVIEW

This supplementary document provides additional implementation details, full per-epoch training curves, parameter breakdowns, complete pre-training diagnostics, and extended mathematical descriptions supporting the main manuscript.

II. COMPLETE PARAMETER COUNTS AND MEMORY OVERHEAD

Table I presents the exact parameter allocation across all four experimental configurations. The frozen backbones (ViT-B/16 and RoBERTa-base) comprise 209.854 M parameters. The newly introduced trainable projection, recurrence, and coupling components require between 0.330 M and 0.422 M parameters ($< 0.21\%$ of total parameters).

TABLE I
DETAILED PARAMETER BREAKDOWN ACROSS EXPERIMENTAL CONFIGURATIONS.

Configuration	Frozen Params (M)	Trainable Params (M)	Total Params (M)	Trainable Ratio (%)
SSD Baseline	209.854	0.330	210.184	0.157%
HEDO-HVSC w/o HEDO	209.854	0.356	210.209	0.169%
HEDO-HVSC w/o HVSC	209.854	0.397	210.251	0.189%
Full HEDO-HVSC (Ours)	209.854	0.422	210.276	0.201%

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III. PRE-TRAINING DIAGNOSTIC AND COORDINATE TRAJECTORIES

Prior to starting full benchmark training, the numerical stability and coordinate-momentum behavior of HEDO were evaluated on initial test batches.

A. Image Feature Transformation

- Raw Input Energy: $\mathcal{H}_{\text{raw}} = 170.824$
- Post-HEDO Energy: $\mathcal{H}_{\text{HEDO}} = 77.563$
- Cosine Similarity: $\text{sim}(\mathbf{X}_{\text{raw}}, \mathbf{X}_{\text{HEDO}}) = 0.9952$
- Unforced Trajectory over 5 steps: $[170.824, 170.651, 170.927, 171.646, 172.801, 174.387]$

B. Text Feature Transformation

- Raw Input Energy: $\mathcal{H}_{\text{raw}} = 5.849$
- Post-HEDO Energy: $\mathcal{H}_{\text{HEDO}} = 77.414$
- Cosine Similarity: $\text{sim}(\mathbf{X}_{\text{raw}}, \mathbf{X}_{\text{HEDO}}) = 0.9920$
- Unforced Trajectory over 5 steps: $[5.849, 5.796, 5.774, 5.783, 5.823, 5.892]$

C. Initial Layer Gradient Norms

- Custom SSD Recurrence: $\|\nabla_{\theta}\|_{\text{SSD}} = 2.3619$
- HEDO Operator: $\|\nabla_{\theta}\|_{\text{HEDO}} = 0.8861$
- HVSC Coupling Block: $\|\nabla_{\theta}\|_{\text{HVSC}} = 1.3847$

IV. DETAILED DATASET SPLIT MANIFEST

The official Flickr8k split was strictly enforced without data leakage:

- **Train Set:** 6,000 unique image filenames, 30,000 text descriptions.
- **Validation Set:** 1,000 unique image filenames, 5,000 text descriptions.
- **Held-Out Test Set:** 1,000 unique image filenames, 5,000 text descriptions.
- $\text{Train} \cap \text{Val} = \emptyset, \text{Train} \cap \text{Test} = \emptyset, \text{Val} \cap \text{Test} = \emptyset$.

V. EMPIRICAL LATENCY SCALING DATA

Table II provides the exact microsecond-level timing measurements across sequence lengths $L \in \{16, 32, 64, 128, 256\}$ on an NVIDIA Tesla T4 GPU ($B = 16, d_{\text{model}} = 128, d_{\text{state}} = 64$).

Linear regression parameters:

$$\text{Latency}(L) = 0.149 \cdot L + 0.582 \text{ ms} \quad (R^2 = 0.9998) \quad (1)$$

TABLE II
EMPIRICAL LATENCY MEASUREMENTS FOR CUSTOM PYTORCH SSD RECURRENT SCAN.

Sequence Length (L)	Measured Latency (ms)	Standard Error (ms)	Linear Model Prediction (ms)
16	2.89356	± 0.021	2.966
32	5.43640	± 0.034	5.350
64	10.31483	± 0.052	10.118
128	19.71283	± 0.089	19.654
256	38.60174	± 0.141	38.726